Week 3 — A High Stakes Investigation

Overview

This week introduced **digital forensics and incident response (DFIR)**. Working through a fictional but realistic scenario, the investigation required reconstructing a security incident using log analysis, file metadata, and timeline correlation — the core skills of a SOC analyst or digital forensic examiner.

Topics Covered

- Introduction to digital forensics methodology
- Evidence handling and chain of custody principles
- Log analysis and correlation
- File metadata examination
- Timeline reconstruction
- Incident reporting

Assignment — "A High Stakes Investigation"

Investigated a simulated security incident at a fictional company. The scenario involved determining whether a breach had occurred, who was responsible, and what data was accessed or exfiltrated. Work included:

- Analyzing system and application logs to identify anomalous activity
- Examining file metadata (creation, modification, access timestamps)
- Correlating events across multiple log sources to build an attack timeline
- Identifying the attacker's entry point, lateral movement, and actions on objectives
- Writing a formal incident investigation report with findings and recommendations

Key Concepts

Forensics Methodology (Order of Volatility)

1. CPU registers and cache
2. Running processes and RAM
3. Network connections
4. Disk and file system data
5. Logs and archived data

Log Analysis Fundamentals

- Identifying failed authentication attempts (brute force indicators)
- Spotting privilege escalation events
- Correlating timestamps across systems
- Recognizing data exfiltration patterns (large outbound transfers, unusual hours)

Incident Report Structure

- Executive Summary
- Incident Timeline
- Technical Findings
- Impact Assessment
- Remediation Recommendations

Tools Referenced

- Linux command-line log analysis (`grep`, `awk`, `sort`, `uniq`)
- File metadata tools (`stat`, `ls -la`)
- Manual log correlation techniques